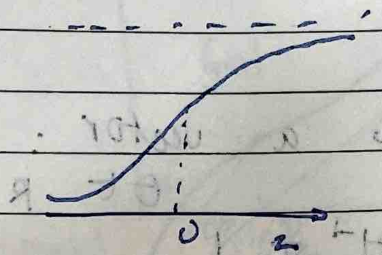
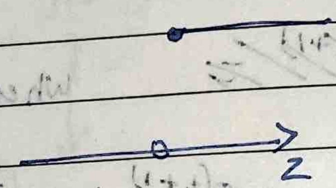
LECTURE-4 PERCEPTRON & GENERALIZED LINEAR MODEL

Topics for today -

- Perceptron
- Exponential Family
- Generalized Linear Models
- Softmax Regression (multiclass classification)

Perceptron





$$g(z) = \frac{1}{1+e^{-z}}$$

$$h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$$

$$g(z) = \begin{cases} 1 & z \geq 0 \\ 0 & z < 0 \end{cases}$$

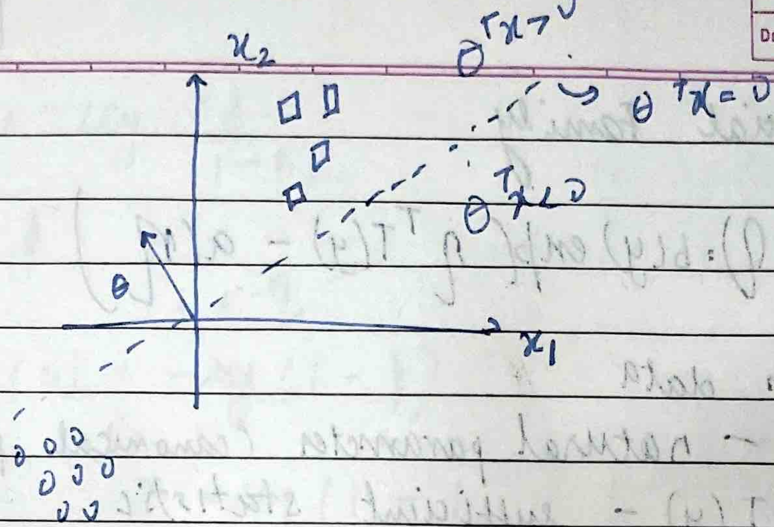
$$h_\theta(x) = g(\theta^T x)$$

$$\theta_j := \theta_j + \alpha\left(y^{(i)} - h_\theta(x^{(i)})\right)x_j^{(i)}$$

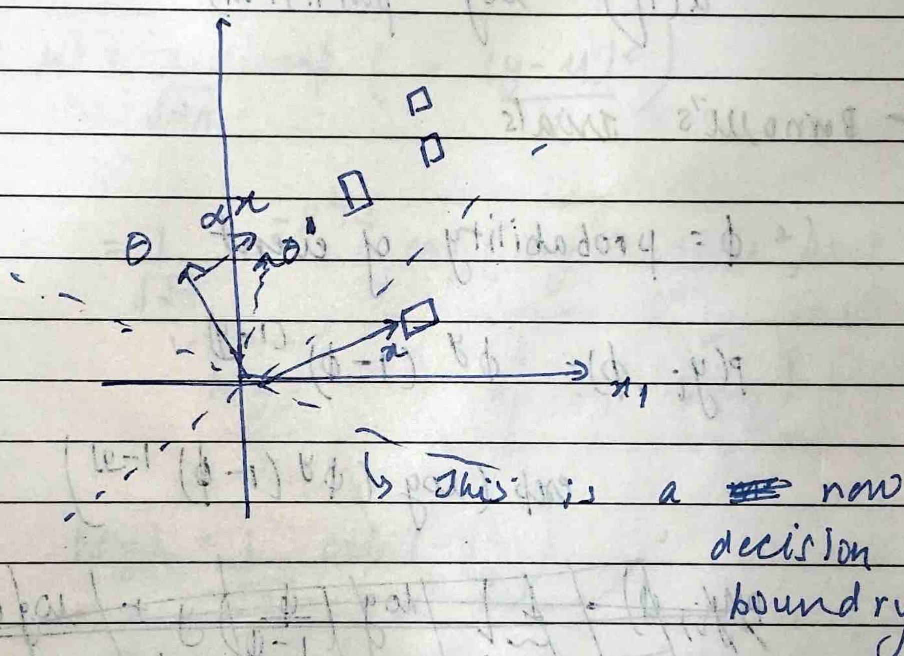
$0 \rightarrow$ if the algorithm got it right

$1 \rightarrow$ if the algorithm got it wrong $y^{(i)} = 1$

$-1 \rightarrow$ if algorithm got it wrong $y^{(i)} = 0$



A vector equivalent of a line is perpendicular to the line.



For $\theta \approx x \mid y = 1$

$\theta \not\approx x \mid y = 0$

Exponential Family

$$p(y;\eta) = b(y)\exp(\eta^T T(y) - a(\eta))$$

$y \rightarrow$ data

η – natural parameter (canonical parameter).

$T(y)$ – sufficient statistic
$\approx y$

$b(y)$ = Base measure

$a(\eta)$ - log partition

- Bernoulli's Trials

ϕ = probability of event

$$p(y_i;\phi) = \phi^y (1-\phi)^{(1-y)}$$

$$= \exp(\log(\phi^y (1-\phi)^{1-y}))$$

$$\cancel{p(y_i;\phi) = \underbrace{1}_{b(y)} \left[\underbrace{\log\left(\frac{\phi}{1-\phi}\right)}_{\eta} \underbrace{y}_{T(y)} + \underbrace{\log(1-\phi)}_{a(\eta)}\right]}$$

$$= 1 \cdot \exp\left[\underbrace{\log\left(\frac{\phi}{1-\phi}\right)}_{\eta} \underbrace{y}_{T(y)} + \underbrace{\log(1-\phi)}_{a(\eta)}\right]$$

$b(y) = 1$

$T(y) = y$

$$\eta = \log\left(\frac{\phi}{1-\phi}\right)$$

$$\phi = \frac{1}{1+e^{-\eta}}$$

$$a(\eta) = -\log(1-\phi)$$

$$= -\log\left(1 - \frac{1}{1+e^{-\eta}}\right) = \log(1+e^{\eta})$$

Gaussian (with fixed variance)

$$p(y;\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y-\mu)^2}{2}\right)$$

$$= \underbrace{\frac{1}{\sqrt{2\pi}} e^{-y^2/2}}_{b(y)} \exp(\underbrace{\mu y}_{\eta\, T(y)} - \underbrace{\tfrac{1}{2}\mu^2}_{a(\eta)})$$

$$b(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right)$$

$$T(y) = y$$

$$\eta = \mu$$

$$a(\eta) = \frac{\mu^2}{2} = \frac{\eta^2}{2}$$

Properties.

a) MLE wrt $\eta \Rightarrow$ concave

NLL is convex

(Negative log likelihood

b) $E[y;\eta] = \frac{\partial}{\partial \eta} a(\eta)$

expectation

mean

c) Variance: $\frac{\partial^2}{\partial \eta^2} a(\eta)$

Generalized Linear Models (GLMs)

Real - Gaussian

Binary - Bernoulli

(non negative integers) Count - Poisson

R^+ - Gamma, exponential

Depending on the data, if we are regression, $y \to$ Gaussian

classification, $y \to$ Bernoulli

Assumptions / Design choices

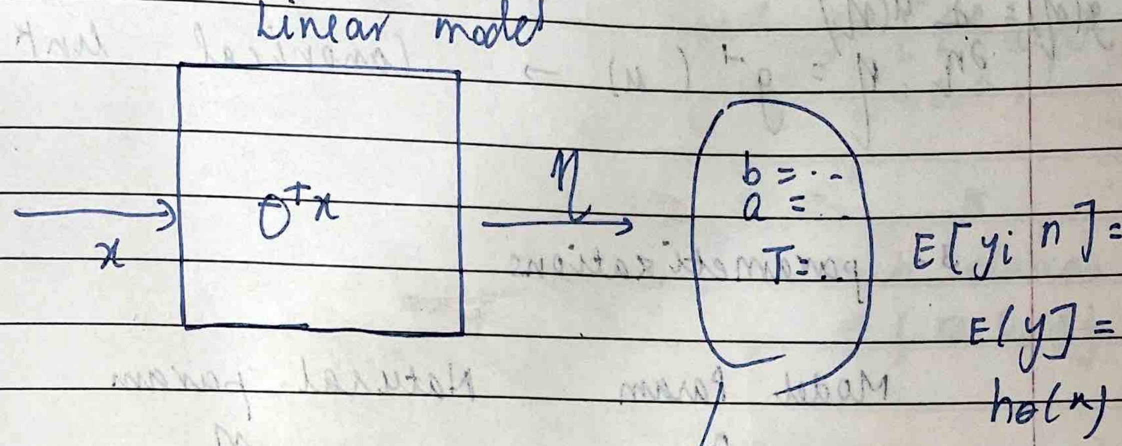
(i) $y \mid x;\theta \sim$ Exponential Family

(ii) $\eta = \theta^T x \quad \theta \in \mathbb{R}^n, x \in \mathbb{R}^n$

(iii) Test time: output will be : $E[y|x;\theta]$

$$h(\theta) = E[y \mid x;\theta]$$

Linear model



Training time
max θ
maximum likelihood
$\log P(y^{(i)}; \theta^T x^{(i)})$

GLM training

Learning update rule

$$\theta_j := \theta_j + \alpha\left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$

for all GLMs, the update rule is same.

Terminology: $\eta \rightarrow$ natural parameter

$\mu = E[y_i; \eta] = g(\eta)$ - canonical response function.

$g(\eta) = \frac{\partial}{\partial \eta} a(\eta)$

$\eta = g^{-1}(\mu) \rightarrow$ canonical link function

3 parametrizations

Model Param	Natural param	Canonical param
θ	η	ϕ - Ber
		$\sigma^2, \mu \rightarrow$ gau
		$\lambda \rightarrow$ poisson

$\uparrow$ Learn

$\theta^T x \rightarrow$ design choice

$g \rightarrow$

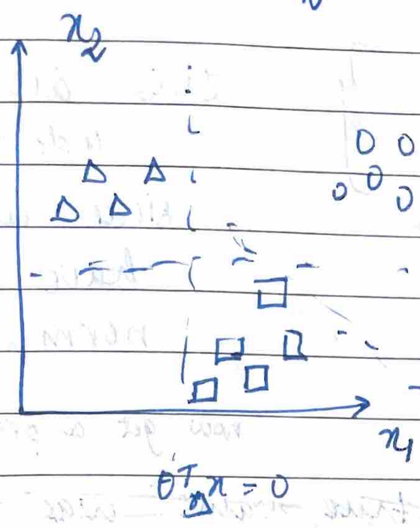
$\leftarrow g^{-1}$

Logistic Regression

$h_\theta(x) = E[y \mid x; \theta] = \phi = \frac{1}{1+e^{-\eta}} = \frac{1}{1+e^{-\theta^T x}}$

Softmax Regression

Cross entropy



$x^{(i)} \in \mathbb{R}$

Label $y = [\{0,1\}^k]$

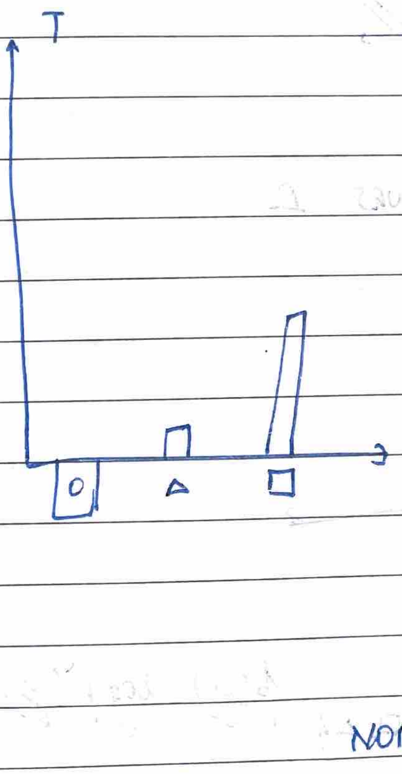
eg. $[0, 0, 0 .., 1 ..]$

$\theta_{class} \in \mathbb{R}^n$ K such classes $\in \{\square, O, \Delta\}$

$$K \begin{bmatrix} - \theta_1 - \\ - \theta_2 - \\ \vdots \end{bmatrix}$$

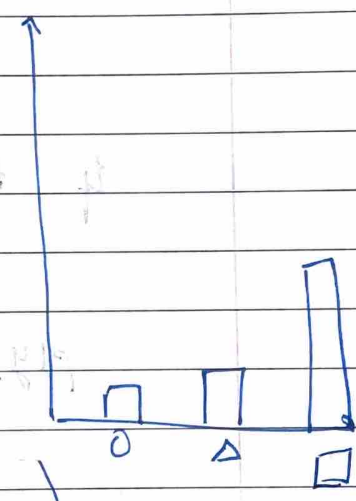
given x $\theta^T x$

logit space



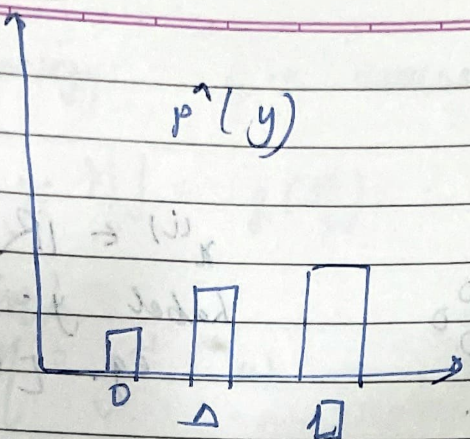
~~exp~~ ⇒

$\exp(\theta^T x)$ ⇒



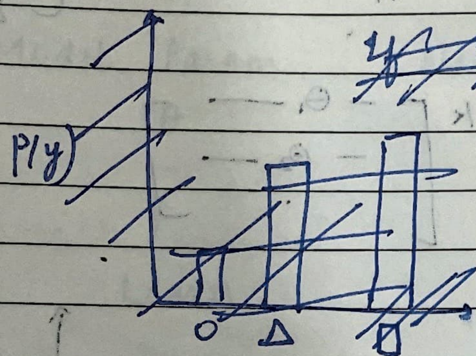
Normalize.

$$\frac{e^{\theta^T x}}{\Sigma\, e^{\theta_i^T x}}$$

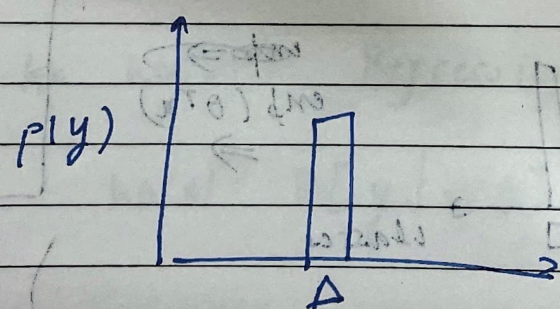


This all adds up to 1 since we have normalized it. We now get a probability distri...

~~If true value was Δ~~



If true value was Δ



$$\text{Cross ent}(p, \hat{p}) = \sum_{y \in \{o, \square, \Delta\}} p(y) \log \hat{p}(y)$$

$$= -\log \hat{p}(y_\Delta)$$

$$= -\log \frac{e^{\theta_\Delta^T x}}{\sum_{c \in \{o, \Delta, \square\}} e^{\theta_c^T x}}$$

Perform gradient descent wrt parameters.